

Indices

Laws of Indices

34. Law 1 of Indices: $a^m \times a^n = a^{m+n}$
Law 2 of Indices: $a^m \div a^n = a^{m-n}$, if $a \neq 0$
Law 3 of Indices: $(a^m)^n = a^{mn}$
Law 4 of Indices: $a^n \times b^n = (a \times b)^n$
Law 5 of Indices: $a^n \div b^n = \left(\frac{a}{b}\right)^n$, if $b \neq 0$

(a) Given that $5^{18} \div 125 = 5^k$, find k .

(b) Simplify $3 \div 6p^{-4}$.

$$\begin{aligned} \text{a)} \quad 5^{18} \div 5^3 &= 5^k \\ 5^{15} &= 5^k \end{aligned}$$

$$k = 15$$

$$\text{b)} \quad 3 \div 6p^{-4}$$

$$= \frac{3}{6p^{-4}}$$

$$= \frac{1}{2p^{-4}}$$

$$= \frac{p^4}{2}$$

Zero Indices

35. If a is a real number and $a \neq 0$, $a^0 = 1$.

Negative Indices

36. If a is a real number and $a \neq 0$, $a^{-n} = \frac{1}{a^n}$.

Fractional Indices

37. If n is a positive integer, $a^{\frac{1}{n}} = \sqrt[n]{a}$, where $a > 0$.

38. If m and n are positive integers, $a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$, where $a > 0$.

Simplify $\frac{3y^2}{5x} \div \frac{3x^2y}{20xy}$ and express your answer in positive indices.

$$\begin{aligned} & \frac{\cancel{3}y^{\cancel{2}}}{\cancel{5}x} \times \frac{\overset{+}{\cancel{20}x^{\cancel{2}}}\cancel{y}}{\cancel{3}x^{\cancel{2}}\cancel{y}} \\ &= \frac{4y^2}{x^2} \end{aligned}$$

(a) Simplify $\left(\frac{x^2}{5}\right)^3 \div \frac{50}{3x^0}$.

$$\frac{x^6}{5^3} \times \frac{3}{50}$$

$$= \frac{3x^6}{6250}$$

Answer: $\frac{3x^6}{6250}$ [2]

(b) Simplify $\left(\frac{(3a)^2 b}{20c^5}\right)^{-1} \times \frac{a^3 b^{-2}}{4c}$.

$$= \left(\frac{20c^5}{(3a)^2 b}\right) \times \frac{a^3}{4b^2 c}$$

$$= \frac{20c^5}{9a^2 b} \times \frac{a^3}{4b^2 c}$$

$$= \frac{5ac^4}{9b^3}$$

Answer: $\frac{5ac^4}{9b^3}$ [2]

(a) Simplify $(-3q^2 r^{-2})^2$, leaving your answer in positive index.

$$= \left(\frac{-3q^2}{r^2}\right)^2$$

$$= \frac{9q^4}{r^4}$$

Answer $\frac{9q^4}{r^4}$ [2]

(b) Given that $7^k = \sqrt{343}$, find the value of k .

$$7^k = \sqrt{7^3}$$

$$7^k = 7^{\frac{3}{2}}$$

$$k = \frac{3}{2}$$

Answer $k = \frac{3}{2}$ [2]

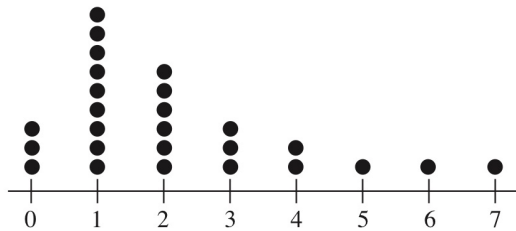
Stats — Data Handling

Dot Diagram

The table shows the number of goals scored by a soccer team during the tournament season.

Number of goals	0	1	2	3	4	5	6	7
Number of matches	3	9	6	3	2	1	1	1

The data can be represented on a dot diagram.



Stem-and-Leaf Diagram

3. In a stem-and-leaf diagram, the stems must be arranged in numerical order and the leaves must be arranged in ascending order.

e.g. The ages of 15 shoppers are as follows:

32	34	13	29	38
36	14	28	37	13
42	24	20	11	25

The data can be represented on a stem-and-leaf diagram.

Stem	Leaf
1	1 3 3 4
2	0 4 5 8 9
3	2 4 6 7 8
4	2

Key: 1 | 3 means 13 years old

The tens are represented as stems and the ones are represented as leaves.

The values of the stems and the leaves are arranged in ascending order.

e.g. The scores for a quiz of two classes are shown in the table.

Class A	55	98	67	84	85	92	75	78	89	64
	72	60	86	91	97	58	63	86	92	74
Class B	56	67	92	50	64	83	84	67	90	83
	68	75	81	93	99	76	87	80	64	58

A back-to-back stem-and-leaf diagram can be constructed based on the given data.

Leaves for Class B	Stem	Leaves for Class A
8 6 0	5	5 8
8 7 7 4 4	6	0 3 4 7
6 5	7	2 4 5 8
7 4 3 3 1 0	8	4 5 6 6 9
9 3 2 0	9	1 2 2 7 8

Key: 58 means 58 marks

Note that the leaves for Class B are arranged in ascending order from the right to the left.

Mean

8. The mean of a set of n numbers $x_1, x_2, x_3, \dots, x_n$ is denoted by $\bar{x}$.

9. For ungrouped data,

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n}.$$

10. For grouped data,

$$\bar{x} = \frac{\sum fx}{\sum f}$$

where f is the frequency of data in each class interval and x is the mid-value of the interval.

Median

11. The median is the value of the middle term of a set of numbers arranged in ascending order.

Given a set of n terms, if n is odd, the median is the $\left(\frac{n+1}{2}\right)^{\text{th}}$ term;

if n is even, the median is the average of the two middle terms.

e.g. Given the set of data: 5, 6, 7, 8, 9.

There is an odd number of data.

Hence, median is 7.

e.g. Given a set of data 5, 6, 7, 8.

There is an even number of data.

Hence, median is 6.5.

Mode

12. The mode of a set of numbers is the number with the highest frequency.

13. If a set of data has two values which occur the most number of times, we say that the distribution is bimodal.

e.g. Given a set of data: 5, 6, 6, 6, 7, 7, 8, 8, 9.

6 occurs the most number of times.

Hence, the mode is 6.

Standard Deviation

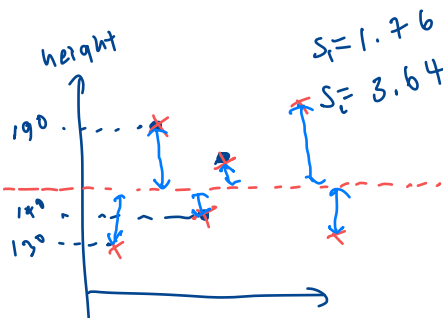
14. The standard deviation, s , measures the spread of a set of data from its mean.

15. For ungrouped data,

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \quad \text{or} \quad s = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}$$

16. For grouped data,

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} \quad \text{or} \quad s = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$



The following set of data shows the number of books borrowed by 20 children during their visit to the library.

0, 2, 4, 3, 1, 1, 2, 0, 3, 1
1, 2, 1, 1, 2, 3, 2, 2, 1, 2

Calculate the standard deviation.

Solution

Represent the set of data in the table below.

x	Number of books borrowed	0	1	2	3	4
y	Number of children	2	7	7	3	1

Standard deviation

$$\begin{aligned}
 &= \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2} \quad x^2 + x^2 + \dots \\
 &= \sqrt{\frac{0^2(2) + 1^2(7) + 2^2(7) + 3^2(3) + 4^2(1)}{20} - \left(\frac{0(2) + 1(7) + 2(7) + 3(3) + 4(1)}{20}\right)^2} \\
 &= \sqrt{\frac{78}{20} - \left(\frac{34}{20}\right)^2} \\
 &= 1.00 \text{ (to 3 s.f.)}
 \end{aligned}$$

The mass, in grams, of 80 stones are given in the table.

Mass (m) in grams	Frequency
$20 < m \leq 30$	20
$30 < m \leq 40$	30
$40 < m \leq 50$	20
$50 < m \leq 60$	10

Find the mean and the standard deviation of the above distribution.

Solution

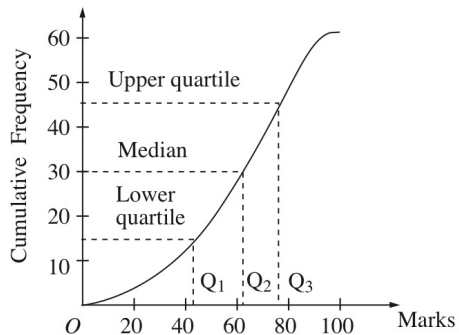
Mid-value (x)	Frequency (f)	fx	fx^2
25	20	500	12 500
35	30	1050	36 750
45	20	900	40 500
55	10	550	30 250

$$\begin{aligned}
 \text{Mean} &= \frac{\sum fx}{\sum f} \\
 &= \frac{500 + 1050 + 900 + 550}{20 + 30 + 20 + 10} \\
 &= \frac{3000}{80} \\
 &= 37.5 \text{ g}
 \end{aligned}$$

$$\begin{aligned}
 \text{Standard deviation} &= \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2} \\
 &= \sqrt{\frac{12\,500 + 36\,750 + 40\,500 + 30\,250}{80} - \left(\frac{3000}{80}\right)^2} \\
 &= \sqrt{1500 - 37.5^2} \\
 &= 9.68 \text{ g (to 3 s.f.)}
 \end{aligned}$$

Cumulative Frequency Curve

17. The following figure shows a cumulative frequency curve.



18. Q_1 is called the lower quartile or the 25th percentile.

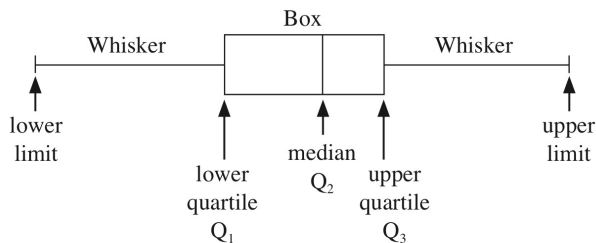
19. Q_2 is called the median or the 50th percentile.

20. Q_3 is called the upper quartile or the 75th percentile.

21. $Q_3 - Q_1$ is called the interquartile range.

Box-and-Whisker Plot

22. The following figure shows a box-and-whisker plot.



23. A box-and-whisker plot illustrates the range, the quartiles and the median of a frequency distribution.

The heights of a class of students are shown in the following stem-and-leaf diagram.

Stem	Height of class
14	1 5 5 6 7 9
15	1 3 3 3 4 4 5 5 5 5
16	0 0 1 1 2 2 4 7

Key: 14 | 1 means 141 cm

(a) Write down

(i) the median height,

$$\begin{aligned} \text{median term} &= \frac{n+1}{2} \\ &= \frac{24+1}{2} \\ &= 12.5 \end{aligned}$$

$$\frac{154+155}{2} = 154.5$$

Answer 154.5 cm [1]

(ii) the modal height.

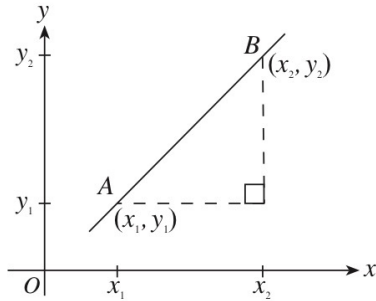
Answer 155 cm [1]

(b) 25% of the students are at least x cm tall.
Find the value of x .

$$24 \times \frac{1}{4} = 6$$

$$6^{\text{th}} \text{ person} = 149$$

Answer $x =$ 149 [2]



Gradient

1. The gradient of the line joining any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ or } \frac{y_1 - y_2}{x_1 - x_2}.$$

2. Parallel lines have the same gradient.

Distance

3. The distance between any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

(a) A is the point $(3, -7)$ and B is the point $(5, 3)$.

(i) Find the length of AB .

$$\begin{aligned} AB &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \\ &= \sqrt{(3 - 5)^2 + (-7 - 3)^2} \\ &= \sqrt{104} \\ &= 10.2(3.s.f) \end{aligned}$$

Answer10.2.....units [2]

(ii) Find the equation of line AB .

$$y = mx + c$$

$$\begin{aligned} m &= \frac{3 - (-7)}{5 - 3} \\ &= 5 \end{aligned}$$

$$y = 5x + c$$

$$\text{sub } x = 3, y = -7$$

$$-7 = 5(3) + c$$

$$c = -22$$

$$y = 5x - 22$$

Answer $y = 5x - 22$ [2]